Exercise 09 - Hash Problems

Problems:

1. Create a cpp file named '`hash01.cpp`' that define the unsigned integer function named `hash()` whose header is

   ```
   size_t hash(char key)
   ```

   that returns the one less than its position in the alphabet if *key* is a letter; otherwise, it returns 26. Afterward, define the Boolean function named `ContainsAll()` whose header is

   ```
   bool ContainsAll(fstream& obj)
   ```

   that returns true only if the file of *obj* opens and contains at least one instance of each letter; otherwise, it returns false. Use direct hashing with a Boolean array.

   Next, within the main function, do:

   a) Create a fstream object.
   b) Prompt the user for a file name.
   c) Open the file for reading.
   d) Display the invocation `ContainsAll()` with the fstream object as its argument.
   e) Close the file.

2. Create a header file named '`PriorityQueue`' that define a hashing priority queue named *PriorityQueue* that contains:

   - Private *Queue* array field named *table* of size 5.
   - Private unsigned integer field named *size*.
   - Private unsigned integer function named `unit()` that takes a constant generic reference parameter and returns 0.
   - Private unsigned integer pointer function named *hash* that takes a constant generic reference parameter.
   - Public special member functions that initializes *hash* to `unit()`.
   - Public overloaded constructor that takes an unsigned integer pointer function parameter that takes a constant generic reference parameter and assigns the parameter to *hash*.
   - Public `enqueue()` method that enqueues into the *table* element based on *hash*.
   - Public `dequeue()` method that dequeues from the first non-empty highest priority *table* element.
   - Public `peek()` method that peeks from the first non-empty highest priority *table* element.
   - Public `empty()` method that returns true if all *table* elements are empty; otherwise, return false.

   Afterward, create a cpp file named '`hash02.cpp`' that includes '`PriorityQueue.h`' and create a task scheduler application.